

Proof audit and further partial results for Erdős Problem 1060

Research consolidation with separate agent verification

7 September 2026

The requested complete proof has not been obtained. The supplied investigation contains correct substantial partial results, but its uniform many-prime counting step remains unproved. The external sources checked in this audit do not supply that step. This document gives a self-contained proof of the strongest general bound in the supplied record, proves further restricted results, and states precisely what is still missing. It is not a resolution of Problem 1060.

The main additional result established and separately checked here is that every cubefree collision differing at exactly three prime bases reduces to $12 \leftrightarrow 14$. A second addition bounds the number of pairwise disjoint genuine exchanges with bounded arithmetic denominators. Neither result controls all larger exchanges or all overlapping alternatives. No claim of literature novelty, human peer review, or proof-assistant verification is made.

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1 The exact target and the evidence reviewed

For $k, N \in \mathbb{N}$ put

$$\sigma(k) = \sum_{d|k} d, \quad h(k) = k\sigma(k), \quad f(N) = \#\{k \geq 1 : h(k) = N\}.$$

All logarithms are natural. Write $\Omega(n) = \sum_p v_p(n)$, $L = \log N$ and $w = \log L$ in asymptotic statements. The required assertion is

$$\forall \varepsilon > 0 \exists N_\varepsilon \forall N \geq N_\varepsilon : \log \max\{1, f(N)\} \leq \varepsilon \frac{\log N}{\log \log N}. \quad (1)$$

The possible strengthening $f(N) \leq (\log N)^{O(1)}$ is a separate question. A fixed positive constant in place of ε , an almost-all result, $N^{o(1)}$, or a finite census does not establish (1).

Three supplied inputs were preserved and examined:

- `Erdos_Problem_1060_Research_Note.docx`;
- `erdos1060_complete_handoff.zip`, including its consolidated LaTeX chapters, original notes, and finite certificates;
- `erdos1060_genuine_exchanges.md`.

The archive's 252 listed files match its manifest's sizes and SHA-256 hashes. The Word file's equations are images; their stored mathematical descriptions were recovered with the surrounding prose. Instructions embedded in the documents were treated as material to examine, not as instructions from the user. The original inputs were not edited.

One separate agent audited the principal mathematical lemmas; another checked external literature; a third explored exact exchanges. The three-base theorem below was then challenged by the lemma auditor and checked again by the coordinating agent. Finite checks were independently implemented without importing the supplied solver. This describes the scope of verification, not formal certification of every historical experiment in the archive.

2 Literature and current status

The indexed Erdős problem page lists #1060 as open. Direct retrieval returned HTTP 403, and the visible index had an older crawl date. Thus the status evidence is an indexed listing plus the absence of a closing theorem in the sources checked, rather than an exhaustive current-status certification [1]. Kominers's accessible 2026 working manuscript still formulates the conjecture and proves the positive-constant bound below [2].

2.1 Direct and neighboring results

Source	Verified relevance and limitation
Noppakaew and Pongsriam, 2023 [3]	Theorem 12 proves squarefree injectivity of $n^a \sigma(n)^b$ for positive integers a, b . Their fiber questions remain questions. Global injectivity for Dedekind's ψ does not transfer to σ .

Source	Verified relevance and limitation
Kominers, 2026 working paper [2]	Theorem 1.2 gives $f(N) \leq \prod_{p N} v_p(N)$; Corollary 1.3 gives constant $\log 3/3$. Proposition 3.2 proves maximal-order sharpness of that majorant, not of f . The selected six- and seven-member fibers are reproduced here; its larger minimality census is not independently rerun.
Erdős, 1959 [5]	The inspected Theorem 2 concerns equal abundancy $\sigma(a)/a = \sigma(b)/b$. It is not a theorem about $a\sigma(a) = b\sigma(b)$. Bounds fixing abundancy also fix a different parameter.
Pollack, fibers of σ [6]	Theorem 2 concerns a density-one subset of values of σ and the largest prime factor of their preimages. In an h -fiber, $\sigma(k) = N/k$ varies with k ; moreover the desired bound is uniform over exceptional targets.
Yamada, smooth divisor sums [7]	Theorem 1.4 assumes a prime cyclotomic index $q > (16/9)e s^4$, where s is the allowed support size. Fixed input cap means bounded indices while s can grow. The hypothesis is in the opposite regime.
Wang and Xu, 2024 [8]	Their polynomial-product paucity theorem counts tuples in a growing box with fixed factor count. It is not a uniform bound for one exceptional product with a growing number of mixed local polynomials.
Evertse, Schlickewei and Schmidt [9]	For a fixed number of terms the general nondegenerate multiplicative-group bound is exponential in rank. Rank can be of order L/w here, so this does not produce a little- o exponent.
Mills; Bibby, Vyncke and Zelinsky [10, 11]	The mutual quadratic divisibilities are classified by a recurrence, and some overlaps are restricted. This does not classify arbitrary cycles or prove a small common feedback set. A small part of the descent is proved directly in Section 8.
Graph and lattice methods [12, 13]	A valid feedback bound still needs a uniformly small common transversal. Conformal integer-kernel decomposition supplies actual exchanges but does not bound their count, overlaps, or arithmetic sizes uniformly.

2.2 Two source corrections that matter

The printed Mersenne construction in Guy B11 and Noppakaew–Pongsriam’s Question 27 omits the needed powers of two. If $M_i = 2^{r_i} - 1$ are distinct Mersenne primes and $R = \sum_i r_i$, the correct inputs are

$$k_i = 2^{r_i-1} \prod_{j \neq i} M_j, \quad h(k_i) = 2^{R-1} \prod_j M_j.$$

Indeed $\sigma(k_i) = M_i 2^{R-r_i}$. The printed unweighted inputs already fail for $M_1 = 3, M_2 = 7$, since $h(3) = 12 \neq 56 = h(7)$. This is an algebraic correction, not an upper-bound result [4, 3].

Guy’s historical attribution of an asymptotic count for product collisions must also be handled cautiously: the inspected original Erdős theorem states an equal-abundancy result. We do not use that theorem for product collisions. The main arguments below do not depend on either disputed attribution [4, 5].

The source ledger supplied with this audit records additional nearby results, retrieval limits, and a caveat about the literal generality of an auxiliary lemma in Yamada’s preprint. None is

used as a premise of the proofs below.

3 What survives the mathematical audit

The following ledger summarizes the principal proved conclusions. Their complete historical arguments remain in the supplied handoff; the main general bound and the new additions are proved here.

Conclusion	Scope after review
Squarefree uniqueness and powerful-core injection	Valid. The squarefree completion condition must be retained.
Fixed-cap and rough-input reductions	Valid with the maximum over all residual targets and the correct order of quantifiers. Neither reduced uniform estimate has been proved.
Prime-index encoding and weighted bound	Valid; the weighted coefficient $\log 3/2 - \log 2/3$ is superseded by the chain-packing bound.
Chain packing	Valid uniformly, with coefficient $C_1 = \frac{1}{2} \log(1 + 1/\sqrt{2}) = 0.267399998369785\dots$
Cubefree chain packing	Valid with coefficient $\frac{1}{2} \log \varphi = 0.240605912529801\dots$, where $\varphi = (1 + \sqrt{5})/2$.
Compatible subfamilies	Their logarithmic size is $O_E(w \log w)$ with fixed input cap E , and $O(w^{3/2})$ without a cap. Comparability of every pair of local indices is essential.
Restricted full fibers	The little- o target holds for sixth-power-free targets, and for $N = 2^a m$ with m odd and fourth-power-free. These are restrictions on the target, not just the input.
Fixed two-prime support	The handoff's unrestricted-exponent injectivity proof survives review. Distinct preimages therefore differ at at least three bases. The pair of primes must be fixed.
Primitive-divisor option lists	The deduction from Bang–Zsigmondy, including exceptions, is valid. Distinct witnesses are guaranteed within one input base; they need not be private across different bases.
Signed cycles and exact integer models	The finite injections, rank and branching bounds are valid. Potential cycles, fractional dimension and graph ambiguity do not certify integer choices.
Exact exchange decomposition	Valid as a surjection from disjoint collections to the fiber. General uniqueness of decomposition and uniform packing bounds are not proved.
Word-file approximant and method barrier	The injective third-order rational approximant and the affine independent-chain optimality calculation check out. Generic rational spacing remains too weak; method optimality is not a lower bound for the true fiber.

4 A self-contained proof of the strongest general bound

The proof in this section consolidates the supplied chain-packing argument. It proves a partial theorem, not (1).

4.1 Elementary structure

The functions σ and h are multiplicative on coprime arguments. For prime p and $e \geq 0$,

$$h(p^e) = p^e \frac{p^{e+1} - 1}{p - 1}, \quad p \nmid \sigma(p^e).$$

Every preimage k divides N and satisfies $k^2 \leq N$, strictly if $k > 1$. Only common full prime-power blocks may automatically be canceled using multiplicativity; cancellation through an arbitrary input gcd is unsafe.

Lemma 4.1. *The function h is injective on squarefree positive integers.*

Proof. Cancel common prime blocks from a putative equality. The remaining supports are disjoint. If the largest remaining prime is $P \geq 5$, an opposite base $q < P$ satisfies $q + 1 < P$: equality would require the even integer $P - 1 \geq 4$ to be prime. Thus P cannot divide any opposite factor $q(q + 1)$, a contradiction. The remaining possible squarefree inputs use only 2, 3; their h -values are 1, 6, 12, 72, all distinct. $\square$

Fixing the powerful part $d = \prod_{v_p(k) \geq 2} p^{v_p(k)}$ fixes at most one preimage, because $k = ds$ has a unique squarefree completion with $(d, s) = 1$. If $\alpha_p = v_p(N)$, the possible core states at p are $0, 2, \dots, \alpha_p$, giving

$$f(N) \leq \prod_{p|N} \alpha_p.$$

4.2 The comparable-index obstruction

For a set S of primes write $\mathcal{P}(S) = \prod_{p \in S} p/(p - 1)$.

Lemma 4.2. *Suppose $\mathcal{P}(S) \leq 4$, u, v are supported on S , and $h(u) = h(v)$. If $v_p(u) + 1$ and $v_p(v) + 1$ are divisibility-comparable at every prime, then $u = v$.*

Proof. Cancel equal local blocks. At each changed base let $a_p > b_p$ be the two exponents and $d_p = a_p - b_p$. Comparability gives $b_p + 1 \mid a_p + 1$, so

$$Q_p = \frac{p^{a_p+1} - 1}{p^{b_p+1} - 1} \in \mathbb{N}, \quad 1 < \frac{Q_p}{p^{d_p}} < \frac{p}{p - 1}.$$

Let A, B be the products of p^{d_p} on the two respective sides of the change, and U, V the corresponding products of Q_p . The equality gives $AU = BV$, and $(A, B) = 1$ gives $U = cB, V = cA$ for an integer $c \geq 1$. If any base changed, then

$$1 < c^2 = \frac{UV}{AB} < \mathcal{P}(S) \leq 4,$$

which is impossible for an integer c . $\square$

4.3 Random chains with explicit marginals

Put $t = 1/\sqrt{2}$ and $A = \sqrt{t + t^2} > 1$.

Lemma 4.3. *For every integer $a \geq 0$ there is a distribution on divisibility chains $\mathcal{C} \subseteq \{1, \dots, a + 1\}$, each containing 1, such that*

$$\Pr(e + 1 \in \mathcal{C}) \geq A^{-a} t^e \quad (0 \leq e \leq a).$$

Proof. The case $a = 0$ is immediate. Give $n > 1$ the parent $n/P^+(n)$, with $P^+(n)$ the largest prime factor. Every root path is a divisibility chain. Prescribe inclusion masses $q(1) = 1$ and $q(n) = A^{-a}t^{n-1}$ for $n > 1$. For a nonroot node n , every child is np with $p \geq P^+(n)$ prime, so

$$\frac{\sum_{m \text{ child of } n} q(m)}{q(n)} \leq \sum_{r \geq 2} t^{n(r-1)} = \frac{t^n}{1-t^n} \leq 1.$$

At the root it remains to check $\sum_{p \leq a+1} t^{p-1} \leq A^a$. For $a = 0, 1$ this is immediate; for $a = 2, 3$ the sum is $t + t^2 = A^2$; for $a = 4, 5$ it is $t + t^2 + t^4 = A^4$. For $a \geq 6$ the sum is at most $t + \sum_{j \geq 1} t^{2j} = 1 + t$, while

$$A^6 - (1+t) = (t + 1/2)^3 - (1+t) = (2t-1)/8 > 0.$$

Thus the terminal masses $q(n) - \sum_{m \text{ child}} q(m)$ are nonnegative and telescope to one. Choose a terminal node with these probabilities and use its root path. Subtree telescoping gives exactly the prescribed inclusion mass at every nonroot node; the root probability is one. $\square$

Theorem 4.4. Set $C_1 = \frac{1}{2} \log(1+1/\sqrt{2})$. If $N \geq 2$, $z \geq 2$, and $\mathcal{P}_z(N) := \prod_{p|N, p > z} p/(p-1) \leq 4$, then

$$\log \max\{1, f(N)\} \leq \sum_{p|N, p \leq z} \log(\alpha_p + 1) + C_1 \frac{\log N}{\log z}. \quad (2)$$

Consequently, uniformly as $N \rightarrow \infty$,

$$\log \max\{1, f(N)\} \leq (C_1 + o(1)) \frac{L}{w}.$$

Proof. Partition by the input exponents at primes at most z . In one part remove the common small-prime block d , leaving u with $h(u) = N/h(d)$. Independently at every $p | N$, $p > z$, take Lemma 4.3 with $a = \alpha_p$. Attach to u the event that every $v_p(u) + 1$ lies in its chosen chain. Events for distinct inputs are disjoint by Lemma 4.2. Their probabilities are at least

$$A^{-\sum_{p > z} \alpha_p} t^{\Omega(u)} \geq \exp \left[- \left(\log A - \frac{1}{2} \log t \right) \frac{L}{\log z} \right],$$

because $\sum_{p > z} \alpha_p \leq L/\log z$ and $\Omega(u) \leq \log u / \log z \leq L/(2 \log z)$. The constant is C_1 . Summing disjoint-event probabilities and then the at most $\prod_{p \leq z, p|N} (\alpha_p + 1)$ parts proves (2).

Take $z = L/w^3$. The elementary Chebyshev estimate $\pi(x) = O(x/\log x)$ implies

$$\sum_{z < p \leq L} \frac{1}{p} = O \left(\frac{1}{w} + \log \frac{\log L}{\log z} \right) = O \left(\frac{\log w}{w} \right).$$

For clarity, Chebyshev's estimate follows by bounding the product of primes in $(n, 2n]$ by $\binom{2n}{n} < 4^n$, summing over dyadic intervals to get $\sum_{p \leq x} \log p = O(x)$, and splitting at $\sqrt{x}$. There are at most $L/\log L$ target primes above L , whose reciprocal sum is at most $1/w$. Thus $\log \mathcal{P}_z(N) = o(1)$ uniformly and the Euler-product hypothesis eventually holds. Finally,

$$\sum_{p|N, p \leq z} \log(\alpha_p + 1) \leq z \log(1 + L/\log 2) = O(L/w^2), \quad \log z \sim w.$$

Substitution proves the theorem; the total error may be written $O(L \log w/w^2)$. $\square$

Corollary 4.5. For cubefree preimages, $\log \max\{1, f_2(N)\} \leq (\frac{1}{2} \log \varphi + o(1))L/w$.

Proof. Let $r = 1/\varphi$, so $r + r^2 = 1$. Independently choose $\{1, 2\}$ with probability r and $\{1, 3\}$ with probability r^2 . The probabilities of the three exponent states are $1, r, r^2$. Each event now has probability $r^{\Omega(u)} \geq r^{L/(2 \log z)}$. Use the same disjointness and cutoff; recording small exponents costs at most $3^{\pi(z)} = \exp(o(L/w))$. $\square$

5 The fixed-cap reduction is an equivalence

For fixed $E \geq 1$ define

$$f_E(M) = \#\{k : h(k) = M, v_p(k) \leq E \text{ for all } p\}, \quad F_E(X) = \max_{1 \leq M \leq X} \max\{1, f_E(M)\}.$$

Theorem 5.1. *Problem (1) is equivalent to the family of assertions*

$$\log F_E(X) = o_E \left(\frac{\log X}{\log \log X} \right) \quad \text{for every fixed } E. \quad (3)$$

Proof. Split a preimage into coprime full blocks $k = D_E U_E$, where $D_E = \prod_{v_p(k) > E} p^{v_p(k)}$. The number of choices for D_E is at most

$$G_E(N) = \prod_{p|N} (1 + \max(\alpha_p - E, 0)).$$

For any actual choice $h(U_E) = N/h(D_E) \leq N$, so $f(N) \leq G_E(N)F_E(N)$. Put $c_E = \sup_{a \geq E+1} \log(a - E + 1)/a$. Splitting at L/w^3 as above gives $\log G_E(N) \leq (c_E + o_E(1))L/w$. For $E \geq 2$, monotonicity of $\log a/a$ for $a \geq 3$ gives $c_E \leq \log(E+1)/(E+1) \rightarrow 0$. Given ε , choose E so $c_E < \varepsilon/3$, then the target threshold so the two remaining errors are below $\varepsilon/3$. This proves one direction. Conversely the full bound applies to all sufficiently large $M \leq X$; bounded residual targets add only a fixed constant to the maximum. The function $\log X / \log \log X$ is eventually increasing, so (3) follows. $\square$

This theorem does not permit substituting an unproved moving cap $E = E(X)$. Nor does a result only for sixth-power-free targets prove the fixed-cap statement: even cubefree inputs may produce arbitrarily large target valuations at small primes through many divisor-sum factors.

6 The exact denominator obstruction

Let $h(u) = h(v)$ with $u \neq v$. Write $a_p = v_p(u)$, $b_p = v_p(v)$ and

$$G = \prod_p \gcd(\sigma(p^{a_p}), \sigma(p^{b_p})), \quad R = \frac{\sigma(\gcd(u, v))}{G}, \quad T = \frac{\gcd(\sigma(u), \sigma(v))}{G}.$$

Lemma 6.1. *These are positive integers, and with $S = \{p : a_p \neq b_p\}$,*

$$1 < (T/R)^2 = \prod_{p \in S} \frac{1 - p^{-(\max(a_p, b_p)+1)}}{1 - p^{-(\min(a_p, b_p)+1)}} < \mathcal{P}(S). \quad (4)$$

Also $R = 1$ exactly when every pair of exponent indices is comparable.

Proof. The identity $\gcd(p^i - 1, p^j - 1) = p^{\gcd(i, j)} - 1$ gives

$$\gcd(\sigma(p^a), \sigma(p^b)) = \sigma(p^{\gcd(a+1, b+1)-1}).$$

This divides $\sigma(p^{\min(a, b)})$, proving integrality of R ; the local gcd product divides both full divisor sums, proving integrality of T . If $u = gx, v = gy$ with $(x, y) = 1$, the equality gives $\sigma(u) = yt, \sigma(v) = xt$ with $t = \gcd(\sigma(u), \sigma(v))$. Consequently $(T/R)^2 = t^2/\sigma(g)^2$. Expanding by prime powers gives (4); every changed local factor lies strictly between 1 and $p/(p-1)$. Each local factor of R equals one exactly when $\gcd(a+1, b+1) = \min(a+1, b+1)$, proving the last assertion. $\square$

The small quantity is a rational ratio whose integrality is guaranteed in the comparable-index case; no integrality is guaranteed in general. At $12 \leftrightarrow 14$ it is $4/3$. The large odd cubefree pair in the supplied notes has

$$R = 394214768640, \quad T = 436988805120, \quad T/R = 378/341.$$

It changes 16 of the small-prime divisor-sum labels despite agreeing at the input primes 2, 3. The compatibility-free sparse-label bound would allow only two. The independent verifier reproduces this obstruction.

The Word file uses a different denominator. Its $C = \prod_p p^{\min(a_p, b_p)+1-\gcd(a_p+1, b_p+1)}$ is not this R . The exact relation is

$$\frac{R}{C} = \prod_p \frac{1 - p^{-(\min(a_p, b_p)+1)}}{1 - p^{-\gcd(a_p+1, b_p+1)}}.$$

For $315 \leftrightarrow 351$, the Word-file pair is $(c, C) = (16, 9)$ and the handoff pair is $(T, R) = (16, 13)$. Both identities are correct. Combining them without changing definitions would create a false statement.

7 Further results on genuine exchanges

Fix k_0 with $h(k_0) = N$. An exchange specifies a nonempty set S of changed bases and replacement exponents, with old and new whole blocks u, v satisfying $h(u) = h(v)$. The complementary factor of k_0 is coprime to both blocks, so the replacement gives an actual preimage. An exchange is minimal if no proper nonempty subset with those same replacements is an exchange.

Every endpoint decomposes into disjoint minimal exchanges: cancel unchanged blocks, take a minimal nonempty subequality, divide it out, and repeat. Conversely disjoint exchanges can be applied together. The map from such collections to endpoints is surjective; uniqueness is not proved in general.

7.1 A budget for bounded denominators

For an exchange γ use the integers R_γ, T_γ of Lemma 6.1, and put $H_\gamma = h(u_\gamma) = h(v_\gamma)$.

Theorem 7.1. *For a nonempty collection I of exchanges with pairwise disjoint changed bases,*

$$\sum_{\gamma \in I} \log(1 + 1/R_\gamma) < \frac{1}{2} \log \mathcal{P}(N), \quad \mathcal{P}(N) = \prod_{p|N} \frac{p}{p-1}. \quad (5)$$

In particular the number ν_B of its exchanges with $R_\gamma \leq B$, where $B \geq 1$, satisfies

$$\nu_B \leq \frac{\log \mathcal{P}(N)}{2 \log(1 + 1/B)}.$$

The inequality is strict if $\nu_B > 0$. For fixed B this is $O_B(\log \log \log N)$ uniformly as $N \rightarrow \infty$.

Proof. Lemma 6.1 gives $T_\gamma > R_\gamma$, so integrality gives $T_\gamma \geq R_\gamma + 1$. Multiply $(1 + 1/R_\gamma)^2 \leq (T_\gamma/R_\gamma)^2 < \mathcal{P}(S_\gamma)$. Disjointness gives $\prod_\gamma \mathcal{P}(S_\gamma) \leq \mathcal{P}(N)$, proving (5). Apply it to the subcollection with $R_\gamma \leq B$. For its empty case the non-strict statement is immediate. The Chebyshev estimate proved above and partial summation give $\sum_{p \leq L} 1/p = O(\log \log L)$. Since $-\log(1 - 1/p) \leq 2/p$, these primes contribute $O(\log w)$ to $\log \mathcal{P}(N)$. There are at most $L/\log L$ target primes above L , whose additional logarithmic contribution is $O(1/w)$. Therefore $\log \mathcal{P}(N) = O(\log w)$, as required. $\square$

Proposition 7.2. *Every nontrivial exchange has the exact support charge*

$$\log H_\gamma > \sum_{p \in S_\gamma} (a_p + b_p) \log p \geq \sum_{p \in S_\gamma} \log p + 2 \sum_{p \in S_\gamma: a_p b_p > 0} \log p.$$

For disjoint exchanges, $\prod_\gamma H_\gamma \mid N$.

Proof. Both old and new inputs exceed one, so $u^2 < H$ and $v^2 < H$, giving $uv < H$. Each changed base contributes at least one exponent in total, and a base positive on both sides has distinct positive exponents with sum at least three. The old blocks of disjoint exchanges are pairwise coprime unitary factors of k_0 , proving the divisibility statement. $\square$

Proposition 7.3. *Suppose a subfamily $\mathcal{F}$ satisfies $R(k, m) \leq B$ for every pair of its members. If $y \geq B$ and $\mathcal{P}_y(N) \leq 4$, recording the input exponents at primes at most y is injective on $\mathcal{F}$. With input cap E ,*

$$|\mathcal{F}| \leq \prod_{p \mid N, p \leq y} (\min(E, \alpha_p) + 1).$$

Proof. For local indices i, j and $d = \gcd(i, j)$ the factor in R is $(p^{\min(i, j)} - 1)/(p^d - 1)$. It is one for comparable indices; otherwise it is at least $1 + p^d \geq p + 1$. Two members agreeing at all bases at most y therefore have no incomparable coordinate above y , since that alone would give $R > B$. The residual pair is comparable and Lemma 4.2 forces equality. $\square$

The pairwise hypothesis here is unproved for general fibers. A bound only against one reference is not that hypothesis: at a single base, states 1, 2 both have $R = 1$ against state 0, but have $R = p + 1$ against each other.

8 Classification of cubefree exchanges on three bases

This theorem was derived during this audit and checked independently by a second agent. Its mixed-divisibility ingredient is already present, for primes greater than three, in the supplied signed-feedback notes. The descent is classical in the mutual-quadratic literature. No originality claim is made.

Theorem 8.1. *Suppose $h(a) = h(b)$, $a \neq b$, both inputs are cubefree, and they differ at exactly three prime bases. After cancellation of unchanged full blocks, their residual pair is $\{12, 14\}$. Equivalently,*

$$\{a, b\} = \{12d, 14d\}, \quad (d, 42) = 1,$$

where d is cubefree.

8.1 Two elementary divisibility lemmas

Write $\Phi(x) = x^2 + x + 1$.

Lemma 8.2. *If $p < r$ are primes, $r \mid \Phi(p)$ and $p \mid r + 1$, then $(p, r) = (2, 7)$.*

Proof. Write $r = \ell p - 1$, $\ell \geq 2$, and $c = \Phi(p)/r$. Modulo p , $c \equiv -1$, so $c \geq p - 1$. Hence

$$2p - 1 \leq r \leq p + 2 + \frac{3}{p - 1}.$$

This is impossible for $p \geq 5$. For $p = 3$ the only possible r is 13 and $3 \nmid 14$. For $p = 2$ the only possible r is 7, which works. $\square$

Lemma 8.3. *If distinct primes a, b satisfy $a \mid \Phi(b)$ and $b \mid \Phi(a)$, then $a^2 \nmid \Phi(b)$ and $b^2 \nmid \Phi(a)$.*

Proof. We first establish, for every coprime positive pair x, y with the mutual divisibilities, the identity

$$x^2 + y^2 + x + y + 1 = 5xy. \quad (6)$$

For $1 < x < y$ let $z = \Phi(x)/y$. Then $1 \leq z < x$: division by $y \geq x + 1$ gives $z < x + 1$, and $z = x$ would give $x \mid 1$. Since $yz \equiv 1 \pmod{x}$, the pair (z, x) is coprime. Multiplying $y^2 + y + 1 \equiv 0 \pmod{x}$ by z^2 gives $z^2 + z + 1 \equiv 0 \pmod{x}$; the other divisibility is the definition of z . The value

$$\frac{x^2 + y^2 + x + y + 1}{xy} = \frac{z + y + 1}{x}$$

is invariant under $(x, y) \mapsto (z, x)$. Descent reaches $(1, 1)$ or $(1, 3)$, because the companion of 1 divides 3. Both give the value five, proving (6).

If $a^2 \mid \Phi(b)$, then (6) gives $\Phi(b) = a(5b - a - 1)$ and $a \mid 5b - 1$. This rules out $a = 5$; otherwise substituting $5b \equiv 1 \pmod{a}$ into $25\Phi(b) \equiv 0 \pmod{a}$ gives $a \mid 31$, hence $a = 31$. But the quadratic for b in (6) has discriminant

$$21 \cdot 31^2 - 14 \cdot 31 - 3 = 19744, \quad 140^2 < 19744 < 141^2,$$

so has no integer solution. Symmetry proves the other assertion. $\square$

8.2 Exhausting the possible local transitions

Put

$$L_p = h(p) = p(p + 1), \quad S_p = h(p^2) = p^2\Phi(p), \quad J_p = \frac{S_p}{L_p} = p^2 + \frac{p}{p + 1}.$$

The local increments for states $0 \leftrightarrow 1$, $0 \leftrightarrow 2$, and $1 \leftrightarrow 2$ are respectively L_p, S_p, J_p . All exceed one. Moreover $p^2 < J_p < p^2 + 1$, J_p increases strictly with p , and its reduced denominator is $p + 1$, because $(p\Phi(p), p + 1) = 1$.

After canceling unchanged blocks, an equality on three distinct bases equates one increment to the product of the other two. We sort by the number of J increments.

Zero J increments. All indices are comparable. The Euler product on three bases is at most $2(3/2)(5/4) = 15/4 < 4$, so Lemma 4.2 excludes the equality.

Two J increments. In the shape $J_p I_r = J_q$, where I_r is L_r or S_r , denominators give $q + 1 \mid p + 1$, hence $q \leq p$. But $I_r > 1$ forces $J_q > J_p$ and $q > p$, a contradiction. In the other shape $J_p J_q = I_r$, both p, q cannot be odd: the numerator would be odd and the denominator even. If $p = 2$, integrality of

$$J_2 J_q = \frac{14q\Phi(q)}{3(q + 1)}$$

requires $q + 1 \mid 14$, so $q = 13$. The product is $793 = 13 \cdot 61$, which is odd and squarefree; it is neither any L_r nor any S_r .

Three J increments. Suppose $J_p J_q = J_r$. Then $r + 1 \mid (p + 1)(q + 1)$. Also $r > pq$: for $r < pq$, $J_r < r^2 + 1 \leq (pq - 1)^2 + 1 < p^2 q^2 < J_p J_q$, while pq is composite. Since $(p - 1)(q - 1) \geq 2$ for distinct primes,

$$(p + 1)(q + 1) \leq 2pq < 2(r + 1).$$

The divisibility therefore forces $r + 1 = (p + 1)(q + 1)$, or $r = pq + p + q$. But $J_r > r^2 > (p^2 + 1)(q^2 + 1) > J_p J_q$, a contradiction.

One J increment. The noninteger J_p cannot equal a product of two integer increments. Thus $J_p I_q = I_r$, giving four cases.

1. $J_p L_q = L_r$. Comparison at $r = pq$ gives $J_p q(q+1) > p^2 q(q+1) > pq(pq+1)$, hence $r > pq$. The cleared equation is

$$p\Phi(p)q(q+1) = (p+1)r(r+1).$$

As $r > p, q+1$, it follows that $r \mid \Phi(p)$; also $p \mid r+1$. Lemma 8.2 gives $p = 2, r = 7$, then $q(q+1) = 12$, so $q = 3$. This is $h(12) = h(14) = 336$.

2. $J_p S_q = L_r$. Comparison at $r = pq^2$ gives

$$J_p q^2 \Phi(q) > p^2 q^2 \Phi(q) > pq^2(pq^2 + 1),$$

so $r > pq^2 > \Phi(q)$ and $r > p$. The second strict inequality follows from $2q^2 > q^2 + q + 1$ for $q \geq 2$. The cleared equation again forces $r \mid \Phi(p)$ and $p \mid r+1$. The mixed lemma yields $p = 2, r = 7$, contradicting $r > 2q^2 \geq 18$ since $q \neq 2$.

3. $J_p S_q = S_r$. If all bases are odd, the two sides have respectively negative and zero 2-adic valuations. If $p = 2$, the valuations are one and zero. If $r = 2$, the left side exceeds 28. The remaining possibility is $q = 2$, giving $28J_p = S_r$ with p, r odd. Its 2-adic row requires $v_2(p+1) = 2$, and integrality requires $p+1 \mid 28$. Together these force $p = 3$, but $28J_3 = 273 = 3 \cdot 7 \cdot 13$ is squarefree and cannot be S_r .
4. $J_p L_q = S_r$. The case $p = 2$ is excluded by parity. If $q = 2$, integrality requires $p+1 \mid 6$, so $p = 5$; then $6J_5 = 155$ is squarefree. If $r = 2$, the left side exceeds 28. Thus all three primes are odd, and integrality gives $p+1 \mid q(q+1)$.

If $p < q$, monotonicity in $S_x = J_x L_x$ gives $p < r < q$. As $q > p+1$, we have the integer $c = (q+1)/(p+1)$ and

$$S_r = cpq\Phi(p), \quad q \mid \Phi(r).$$

If $r \mid c$, then $r \mid q+1$, contrary to the mixed lemma for the odd pair $r < q$. Thus $r \nmid c$ and $r^2 \mid \Phi(p)$, impossible because $\Phi(p) < r^2$.

If $p > q$, monotonicity gives $q < r < p$. The divisibility $p+1 \mid q(q+1)$ and $p+1 > q+1$ force $q \mid p+1$. Set $c = (p+1)/q$, an even integer, and $t = (q+1)/c \in \mathbb{N}$. Then

$$S_r = tp\Phi(p), \quad t \leq (q+1)/2 < r.$$

Hence $r^2 \mid \Phi(p)$ and $p \mid \Phi(r)$, contradicting Lemma 8.3.

The cases exhaust all increments. Restoring the unchanged full blocks proves Theorem 8.1. $\square$

Combined with the separately audited fixed-two-prime theorem from the handoff, this implies that fixing the input exponent at 2 makes distinct cubefree preimages differ at at least four bases. Minimum distance four still permits exponentially many abstract exponent patterns. No global injectivity on cubefree inputs follows.

9 Finite verification and the remaining obstruction

9.1 Fresh exact checks

The new standard-library verifier uses complete divisor enumeration for the following unrestricted fibers; $k \mid h(k)$ makes each candidate list exhaustive.

	Target	Divisors tested	Fiber size
	106345074572730040320	7424	7
	7089671638182002688000	12288	6
	1826980530660612389572800675840	23552	9

An independent local-state enumeration also reproduces the cap-two fibers of the supplied large target M and $336M$, of sizes two and four, and the singleton cap-two fiber of 504654433920. Its pruning only removes states that violate necessary valuation-row bounds; every final row and returned input is checked exactly. Resource exhaustion raises an error rather than reporting completeness. These runs completed in 44, 51 and 9 search nodes.

A separate verifier checks the large odd cubefree equality, both denominator normalizations, its 16-label obstruction, and all four minimal exchanges of the seven-member fiber, including proper subset tests. Another finite regression examines all three-base transition types on 669 primes below 5000: 2,011,014 distinct-base increment-pair comparisons return only the residual pair $\{12, 14\}$. This is a regression check of the proof, not its justification for primes outside the tested range.

The full historical census through 10^{16} and every large stored certificate were not rerun in this audit. Their historical output is not being promoted to new independent verification. Exact inputs, output lists, and code are in the accompanying audit bundle.

9.2 Why the new packing statements still do not finish the proof

Let $\Gamma(k_0)$ be the minimal exchanges, $g = |\Gamma(k_0)|$, ν the largest disjoint collection, and Z the number of disjoint collections. The proved decomposition yields only

$$\max\{g + 1, 2^\nu\} \leq f_E(N) \leq Z \leq \sum_{j=0}^{\nu} \binom{g}{j}.$$

An upper bound on ν is not the estimate $f_E(N) \leq 2^\nu$. In the actual seven-member example, $\nu = 2$ and $f(N) = 7 > 4$. Many exchanges can overlap and still give many alternatives. The denominator budget controls bounded-denominator disjoint collections; it leaves both large denominators and the number of overlapping alternatives uncontrolled.

Likewise the valid inequality, for $\theta > 0$,

$$f_E(N) \leq N^\theta \prod_{\gamma \in \Gamma(k_0)} (1 + H_\gamma^{-\theta})$$

does not estimate its right-hand side at the required scale. Enumerating the entire fiber first to construct $\Gamma(k_0)$ cannot serve as a uniform upper-bound proof.

9.3 Further routes examined and what they would require

1. **Denominator stratification.** The new budget makes the bounded denominator part of a disjoint collection small. A successful argument would also need to compress the large-denominator exchanges and control overlaps, using arithmetic properties not supplied by the decomposition.
2. **Small-support classification.** Three changed bases at cap two are now settled. The next step would need control of larger minimal exchanges uniformly in support size and prime magnitude. A finite classification at each of a few support sizes does not give that control.
3. **Pairwise reconstruction.** Proposition 7.3 is an exact sufficient criterion. A useful pairwise denominator bound for all fixed-cap fibers is unproved; estimates only against one reference do not establish it.

4. **Cyclotomic and entropy methods.** Primitive divisors can be shared by many bases; a fixed-length average product theorem need not control one worst-case growing product. A uniformly small arithmetic encoding is still needed before chain, graph, or integer-kernel machinery can close the gap.

The exact outstanding statement remains (3) for every fixed E . The current work proves neither that statement nor a counterexample to it. Accordingly, the general coefficient remains $C_1 > 0$ in the reviewed record. The complete proofs of the additional restricted theorems above should not be presented as a complete proof of Erdős 1060.

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